

# RailPlot : Powerful Data Analytics for Indian Railways

Real-time insights to improve safety and efficiency for Indian railways.

 <https://railplot.com/>

# Problem Statement

## [1] Unutilized Speedometer Data

Although speedometers are installed in all locomotives, **Indian Railways lacks a robust system to analyze the collected speed data** for actionable insights.

## [2] No Automated Violation Detection

There is **no mechanism to automatically detect** violations of speed-related guidelines such as Permanent or Temporary Speed Restrictions from the recorded data.

## [3] Lack of Insight into Loco Pilot Performance

Without proper data analysis, **identifying loco pilot deficiencies**—such as non-compliance, delayed braking, or abrupt handling—is not possible.

## [4] Training and Motivation Gaps

Indian Railways currently lacks **evidence-based feedback loops** to determine whether a loco pilot needs additional **training, counseling, or motivational support** based on driving patterns.

# Proposed Solution

## Visual Insight

Provide clear visual insights through charts and dashboards to help railway management quickly understand trends, identify problem areas, and make informed decisions.

## Violation Detection

Detect violations of PSR/TSR guidelines and flag them.

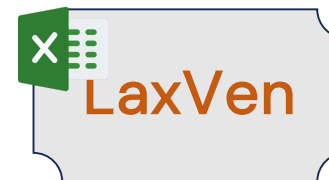
## Performance Reports

Generate loco pilot-specific performance reports to identify whether issues stem from lack of skill, negligence, or the need for motivation.

# Supports All Major Speedometer

Indian railway mainly works with four speedo meters.

Different data file format – One solution.  
RailPlot deals with all kinds of Speedometers.



# Insight Report

|                              |                                                                 |
|------------------------------|-----------------------------------------------------------------|
| From                         | GMO                                                             |
| To                           | ASN                                                             |
| PSR Violation                | STN Station Limit(30)<br>28/05/25 17:04:39 to 28/05/25 17:04:58 |
| TSR Violation                | Nil                                                             |
| Attacking<br>Speed Violation | Achieved Speed: 25.0 Kmph, Req: 50<br>Kmph<br>28/05/25 16:33:46 |



# Dashboard View



**EASTERN RAILWAY, ASANSOL DIVISION**

Ministry of Railways, Govt of India.

Dashboard



N-ASN

Analized By : ARUN KR-1

Departure Time : 18/05/2025 06:10:20

Arrival Time : 18/05/2025 08:40:36

Average Speed : 42.38

Total Distance (KM) : 106.139

PSR Violation : 1

TSR Violation : Nil

Attacking Speed Violation : Nil



View



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[439] BWN-ASN

Analized By :

From: DGR

To : UDL

Departure Time : 18/05/2025 14:32:09

Arrival Time : 18/05/2025 15:40:51

Average Speed : 14.69

Total Distance (KM) : 16.817

PSR Violation : Nil

TSR Violation : Nil

Attacking Speed Violation : Nil



Delete



View



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[438] BWN-ASN

Analized By : ARUN KR-1

From: BWN

To : UDL

Departure Time : 18/05/2025 20:20:00

Arrival Time : 18/05/2025 23:55:01

Average Speed : 22.69

Total Distance (KM) : 81.316

PSR Violation : Nil

TSR Violation : Nil

Attacking Speed Violation : Nil



Delete



View



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[437] BWN-ASN

Analized By : AJITKS

From: BWN

To : ASN

Departure Time : 19/05/2025 08:15:50

Arrival Time : 19/05/2025 10:57:08

Average Speed : 39.37

Total Distance (KM) : 105.832

PSR Violation : Nil

TSR Violation : Nil

Attacking Speed Violation : Nil



Delete



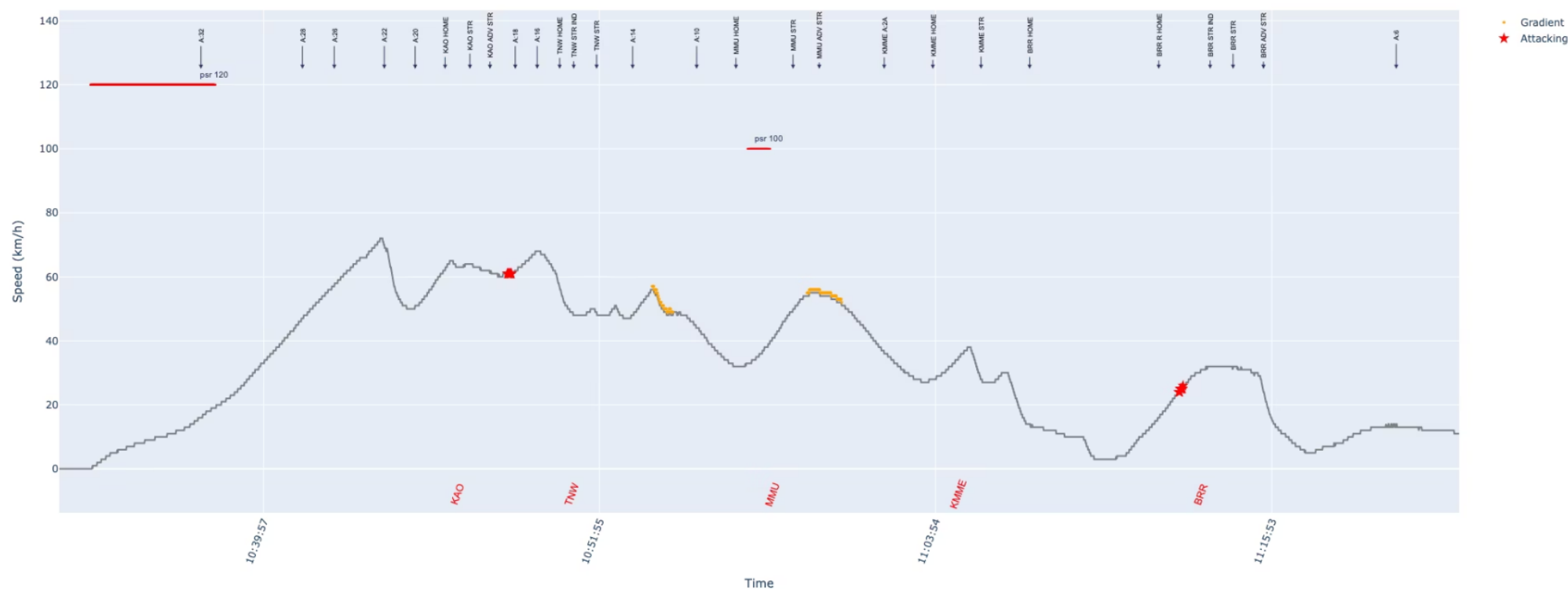
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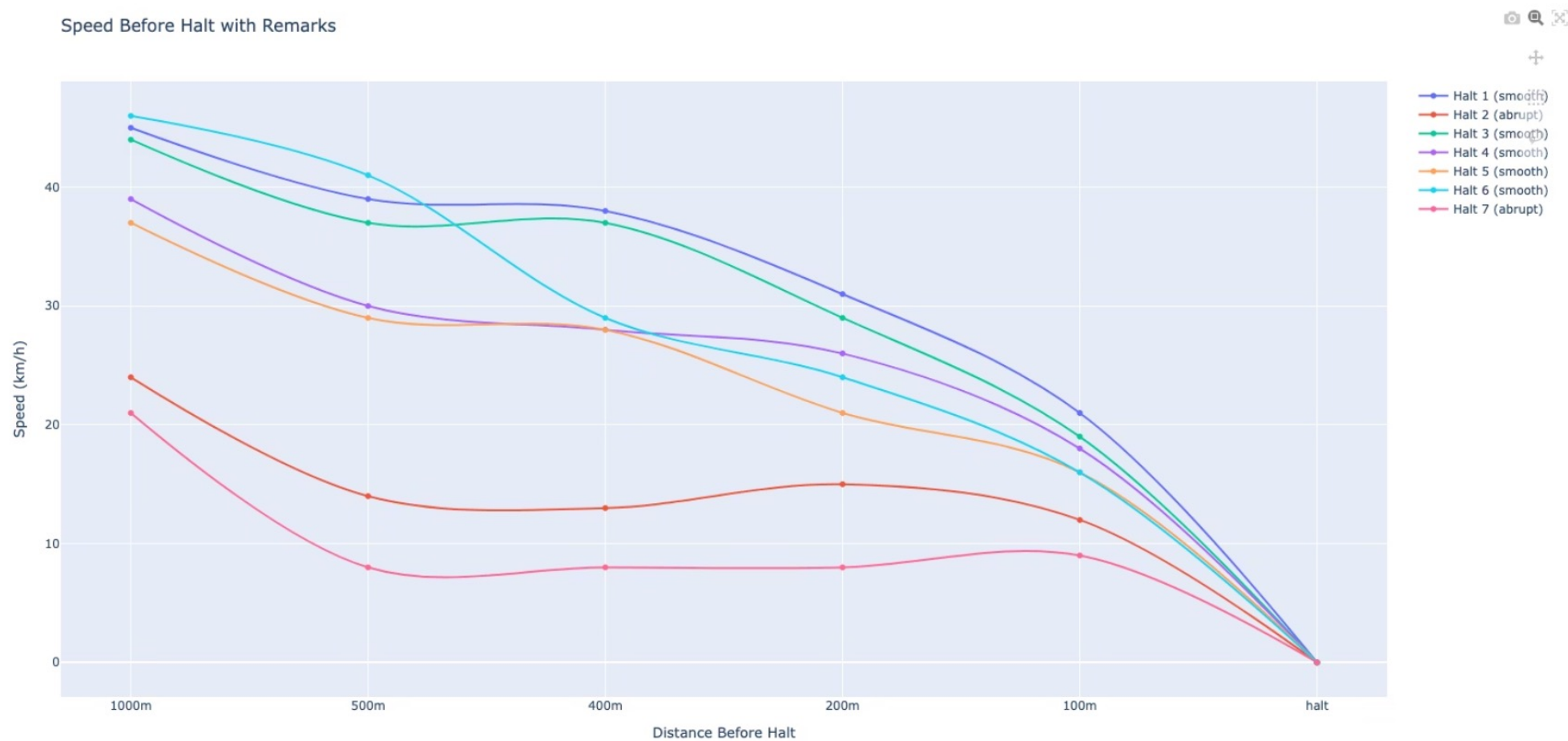
# Speed-Time Chart

Time vs Speed with PSR Speed Sections



# Breaking Profile Analysis

Speed Before Halt with Remarks





# Guideline Violation Detection

## PSR Violation

**ASN PF Line(30)**

22/05/25 23:56:59 To 22/05/25 23:57:50

**STN Station limit(30)**

23/05/25 00:56:39 To 23/05/25 00:56:56

## TSR Violation

NIL

## Attacking Speed Violation

**Achieved Speed: 22 Kmph, Req: 30 Kmph**

23/05/25 01:06:06

## Speed Violation

Pin-point time and location of speed violation.

## Attacking Speed Violation

Identify attacking speed violation for goods train.

## Compliance Enforcement

Prevent accidents by ensuring guideline adherence.

# Insightful Speed-Time Curve Comparison

## Driving Pattern Suggestions

- ★ **Optimized Speed Control**  
Encourages steady acceleration to save energy.
- ★ **Timely Braking**  
Promotes early and smooth braking for safety.
- ★ **Compliance Reinforcement**  
Alerts on deviations to ensure guideline adherence.

### Before Training

Irregular speed with sharp accelerations and braking.

### After Training

Smoother driving with optimal speed control.



# Summary & Impact

Data analytics guide loco pilots for safer, efficient operations.

Actionable insights help reduce accidents and improve fuel economy

Enhanced training leads to smoother driving and passenger comfort.

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